

ZTQ Solutions: Azure Compute Service Strategy

Evaluate workload requirements and select the optimal target Azure compute service

Instructions for Learners

Review each of the three ZTQ Solutions workload scenarios below. Analyze their operational characteristics, technology stack, traffic patterns, and deployment requirements. Based on your evaluation, decide which Azure compute hosting model—**Azure App Service**, **Azure Container Apps**, or **Azure Virtual Machines**—best matches each workload's strategy.

Workload 1: Public-Facing Marketing Platform

SCENARIO 01

Tech Stack & Architecture: Node.js / React stack serving dynamic APIs and mixed static content.

Traffic Pattern: Variable, unpredictable traffic spikes driven by sudden marketing campaign launches.

Key Requirements: High uptime availability, zero management overhead for underlying OS patching/maintenance, and seamless zero-latency deployment swaps (blue-green) with staging validation.

Select Recommended Compute Service Strategy:

☐ Azure Virtual Machines (IaaS) ☐ Azure App Service (PaaS) ☐ Azure Container Apps (ACA)

Workload 2: Recruitment Portal Microservices

SCENARIO 02

Tech Stack & Architecture: Decoupled microservices architecture using containerized .NET Core and Go APIs (handling resume parsing, job postings, and notifications).

Traffic Pattern: Event-driven, asynchronous background tasks that fluctuate heavily based on applicant submission volume.

Key Requirements: Ability to scale microservices independently on event queues, scale down to 0 replicas when idle to minimize cost, and execute progressive canary traffic-splitting without managing underlying Kubernetes clusters.

Select Recommended Compute Service Strategy:

☐ Azure Virtual Machines (IaaS) ☐ Azure App Service (PaaS) ☐ Azure Container Apps (ACA)

Workload 3: Legacy Learning Management System (LMS)

SCENARIO 03

Tech Stack & Architecture: Monolithic course engine requiring custom IIS configurations, Windows Server software dependencies, media transcoding modules, and legacy C++/SQL Server Agent jobs.

Traffic Pattern: Continuous long-running batch processing and video media transcoding.

Key Requirements: Full administrative OS-level control (registry modifications, custom drivers, system agent installations) combined with automated rolling upgrade capabilities across instances.

Select Recommended Compute Service Strategy:

☐ Azure Virtual Machines (IaaS) ☐ Azure App Service (PaaS) ☐ Azure Container Apps (ACA)